

Assessment of task-specific movement patterns to identify ergonomic risks and support injury prevention strategies.

Task:	Production Tote Transfer 1000 line b
Company:	Oxford Frozen Foods Ltd.
Site / Location:	Production Tote Transfer 1000 line b - eae64
Assessment Date:	Not specified
Assessment Method:	REBA (Rapid Entire Body Assessment)
Video Duration:	00:00:11
Assessor:	Vergo Ergonomics Team

How to read this report

REBA, or Rapid Entire Body Assessment, is a screening tool used to assess postural exposure across the neck, trunk, legs, upper arms, lower arms, wrists, force/load, coupling, and activity. The score helps identify whether further ergonomic investigation or task changes may be needed. REBA is not an injury prediction tool. It should be interpreted together with the video context, task frequency, duration, force, workstation layout, and worker feedback.

Section 1 – Assessment Overview

This assessment reviews a production tote transfer task, as observed in a short video sequence lasting approximately 11 seconds. The task involves moving totes along a production line, requiring the worker to reach, grasp, and transfer containers between surfaces or conveyors. The assessment was conducted using the REBA (Rapid Entire Body Assessment) method, which evaluates postural risk factors for the entire body, with a focus on trunk, upper limb, and lower limb mechanics.

A total of 12 representative frames were analyzed, capturing key moments throughout the tote transfer cycle. The selected frames reflect a range of postures, including reaching, grasping, and placing actions. The assessment aims to identify postural risk exposures, highlight modifiable factors, and provide practical recommendations for reducing cumulative strain.

The overall risk level for this task falls within the Medium band, driven primarily by non-neutral wrist positions, extended lower arm postures, and moderate neck flexion. The findings should be interpreted as a screening-level review, as the short video duration may not capture all task variations or cumulative exposures. No significant artefacts or data-quality issues were observed in the analyzed frames.

Section 2 – Summary of Assessment Results

Total Frames Analyzed	12
Average REBA Score	3.7
Overall Risk Level	Medium

Score Distribution

Score Range	Risk Band	Frames (%)	Interpretation
1–3	Low	17% (2)	A minority of frames fell within the Low risk band, typically at the start and end of the task cycle when postures were more neutral.
4–7	Medium	83% (10)	The majority of frames were scored in the Medium risk band, reflecting consistent non-neutral wrist and lower arm postures during tote handling.
8–10	High	0% (0)	No frames reached the High risk band, indicating that extreme postural exposures were not observed.
11–15	Very High	0% (0)	No frames were scored in the Very High risk band.

Interpretation

The average REBA score of 3.67 places this task in the Medium risk band, suggesting that further review and practical improvements are warranted. Most frames showed consistent postural exposures, particularly in the wrists and lower arms, with some variation in neck position. While the risk level is not extreme, addressing these exposures may help reduce cumulative strain, especially if the task is performed frequently or for extended periods.

Section 3 – Task-Based Risk Exposure Analysis

Wrist Deviation and Non-Neutral Hand Posture

Throughout the tote transfer cycle, both wrists—especially the left—are frequently observed in extended or deviated positions, with angles often exceeding the optimal range. This exposure is driven by the need to grasp and maneuver totes at varying heights and orientations, requiring the worker to flex or extend the wrists beyond neutral. Such postures are modifiable through changes in tote placement, handle design, or work height. Sustained or repeated wrist deviation is a recognized risk factor for cumulative strain injuries, making this a priority area for intervention.

Extended Lower Arm and Elbow Posture

The lower arms are consistently held in extended positions, particularly during reaching and placing phases of the tote transfer. Angles frequently fall outside the optimal 60–100 degree range, indicating that the elbows are often straightened. This is inherent to the task setup, as the distance between the worker and the tote or conveyor requires extended reaches. While some extension is unavoidable, reducing reach distance or adjusting workstation layout could help bring the elbows into a more neutral, supported posture, thereby decreasing upper limb loading.

Neck Flexion and Head Position

Moderate neck flexion is observed in several frames, with the worker looking downward to monitor tote placement or alignment. This exposure is driven by the need for visual guidance during the transfer process and is partially inherent to the task. However, excessive or sustained neck flexion can contribute to discomfort or fatigue over time. Improving tote visibility or adjusting work height may help reduce the degree and duration of neck flexion required.

Trunk and Leg Posture

Trunk angles remain within a relatively neutral range throughout the task, with only minor flexion observed. The legs are also maintained in stable, low-risk positions, indicating that the task does not require significant bending, twisting, or unstable stances. These findings suggest that trunk and lower limb exposures are not primary concerns for this task, and current workstation height and floor conditions appear adequate. Maintaining these neutral postures will help minimize cumulative strain on the lower back and legs.

Section 4 – Overall Observations

The tote transfer task is characterized by moderate reach distances and frequent hand repositioning, which drive the observed wrist and lower arm exposures. Work height and tote

placement appear to be set at a level that minimizes trunk flexion and lower limb loading, which is a positive aspect of the current setup.

The main ergonomic concerns relate to upper limb posture, particularly wrist deviation and extended elbow positions during reaching and handling. These exposures are modifiable through workstation adjustments, tote design, or workflow changes. Overall, the task design supports neutral trunk and leg postures, but opportunities exist to further reduce upper limb strain.

Section 5 – Overall Recommendations

Reduce Wrist Deviation by Adjusting Tote Placement

Consider repositioning totes or modifying their orientation to allow for more neutral wrist postures during grasping and transfer. This change can reduce cumulative strain on the wrists and may be implemented by adjusting shelf height, tote handles, or the angle at which totes are presented to the worker.

Minimize Reach Distance to Support Elbow Posture

Evaluate the layout of the workstation to bring totes closer to the worker, reducing the need for extended arm reaches. Shortening reach distance helps maintain elbows in a more neutral, supported position, decreasing upper limb loading and fatigue.

Optimize Work Height to Reduce Neck Flexion

Adjust the height of the work surface or tote presentation area to minimize the need for downward neck flexion during transfer. Improved work height can help keep the head and neck in a more neutral position, reducing the risk of discomfort over time.

Maintain Stable Trunk and Leg Posture

Continue to ensure that the workstation allows for upright trunk posture and stable leg positioning. Regularly review floor conditions and workstation height to support these neutral postures, as they help prevent lower back and leg strain.

Section 6 – Targeted Vergo Training Videos

Module 1: Warm-Up

Relevant as general preparation for repetitive or sustained movement exposure.

Module 14: Using Handheld Devices

Relevant to repetitive handheld tool use, wrist positioning, grip, and hand/arm posture.

Module 2: Power Stance & Power Zone

Relevant to keeping work close to the body and maintaining stronger, more neutral working postures.

Section 7 – Disclaimer

This movement analysis provides general guidance based on the video provided at the time of review. It is not intended to replace medical advice, diagnosis, or professional ergonomic evaluation. Postural risk levels may vary depending on actual workplace conditions and individual worker factors. Employers remain responsible for complying with the Nova Scotia Occupational Health and Safety Act, related regulations, CSA Z1004-12, and all applicable standards.

Vergo's motion-analysis results support proactive identification of movement-related risks but are not a substitute for assessments conducted by certified ergonomists or occupational health professionals. Any images or video content included must be used in accordance with organizational privacy and data-protection policies.

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